

Emigration - a population-migration mod for Civilization VII

Contents

Emigration - a population-migration mod for Civilization VII	2
At a glance (for players)	2
System Guide and Feature Reference	3
Contents	3
1. What it does (player-facing)	3
2. How it works: the per-turn loop	3
3. The signals & the Prosperity score	4
4. Population scaling (Demographics alignment)	5
5. The advanced model (algorithms & per-civ tuning)	5
6. Interactive systems (on by default)	6
7. Consequences: the gameplay-write cost layer (<code>emigration-effects.js</code>)	9
8. Reporting & Demographics integration (<code>emigration-migration-stats.js</code>)	9
9. In-game feedback & notifications (<code>emigration-feedback.js</code> , <code>emigration-events.js</code> , <code>emigration-naming.js</code>)	10
10. Options & tuning	10
11. Architecture / module map	11
12. The execution model & what the probe confirmed	12
13. Persistence	12
14. Localization	13
15. Install & run	13
16. Development	13
17. Caveats & limits	14
18. Compatibility & mod coexistence	14
Credits & license	15

Emigration - a population-migration mod for Civilization VII

At a glance (for players)

Adds real migration and refugee systems to Civilization VII. **Unhappy, poor, besieged, or disaster-struck cities lose population; prosperous and welcoming cities attract it** — within and across civilizations. Immigration brings growth, costs, politics, and Demographics graphs.

- Refugees flee wars and disasters, fleeing *away* from the invader.
- Unhappy, low-yield cities steadily lose population over time, even at peace.
- A **Pro-Immigration Stance** policy attracts migrants (and earns Influence).
- An **Anti-Immigration Stance** policy retains your people and boosts Production, at the cost of Influence.
- Receiving migrants creates a temporary, real assimilation cost — growth isn't free.
- **It tells you why.** Population changes are explained in the moment (a toast: cause, what to do, whether it's temporary, who pays) and on demand per city (`emigration.city(id)`), with a full **Migration dashboard** (`emigration.window()`).
- Fully integrated with the **Demographics** mod: net migration / emigration / immigration / refugees graphs with per-cause source breakdowns, plus a dedicated **Migration** page.

Everything below is the full system reference. Every advanced layer can be tuned or switched off in **Options** → **Mods** → **Emigration - Advanced** (§10), so you can run anything from the plain baseline to the complete model. New to the mod? The bullets above are all you need to start; treat the rest as the manual.

Citizens leave unhappy or struggling settlements including those impacted by environmental disasters and conflict. They move toward happier, more prosperous ones, both **within and between civilizations**.

This mod is driven by a Civ V-style *Prosperity* model.

Migration counts can be reported in **both** the game's own population points (1, 2, 3 ...) **and** a historically representative people count (thousands ... hundreds of millions) scaled to match the **Demographics** mod — shown together by default (e.g. *1 population point (12 thousand people)*), or either alone, via an Options toggle (§10). Absorbing migrants carries a **real, duration-based in-game cost** so growth isn't free.

Several layers sit on top of that baseline, **all on by default**, so you get the full system out of the box. Each can be switched off individually in Options if you'd rather run closer to the bare baseline:

- an advanced Prosperity model (§5): reshapes the happiness draw, time-gates and caps, war displacement, brakes runaway magnets, discounts tall play, and adds bounded per-leader/per-civ tuning;
- interactive systems (§6): refugees who avoid the aggressor that attacked them; Pro/Anti-Immigration stance policies (and base-game Open Borders agreements) that shape cross-civ migration; asylum/relationship permeability; environmental disasters (floods, volcanoes, plague, the last carried by migrants); and an outlet so a trapped, dying population isn't bottled up forever;
- in-game feedback (§9): styled toasts, named refugee headlines, world-news for major crises (spam-throttled), and a Refugees metric with graphs in the Demographics mod;
- localization across all 10 languages (§14).

It all runs in the UI VM (GameFace JS) each turn. This is not a UI-only mod: population moves and yield/Influence changes are real gameplay writes (§12).

Documentation

- Design notes & prior art: `../emigration-docs/DESIGN.md`
- Feasibility / probe evidence: `../emigration-docs/FINDINGS.md`
- In-game testing requirements (what still needs live verification): `../emigration-docs/testing-requirements.md`
- Civ VII modding mechanics & limits: `../emigration-docs/civ7-mechanics-and-feasibility.md`
- How leader/civ abilities & mementos interact (snowballs, exploits): `../emigration-docs/leader-civ-memento-interactions.md`
- The advanced-model design & math (Algorithms A-D + the civ table): `../emigration-docs/algorithmic-improvements.md`

- The interactive-systems design + the build-ready implementation map: ../emigration-docs/interactive-extensions-design.md, ../emigration-docs/interactive-extensions-implementation.md
-

System Guide and Feature Reference

Contents

1. What it does (player-facing)
 2. How it works: the per-turn loop
 3. The signals & the Prosperity score
 4. Population scaling (Demographics alignment)
 5. The advanced model (algorithms & per-civ tuning)
 6. Interactive systems (on by default)
 7. Consequences: the gameplay-write cost layer
 8. Reporting & Demographics integration
 9. In-game feedback & notifications
 10. Options & tuning
 11. Architecture / module map
 12. The execution model & what the probe confirmed
 13. Persistence
 14. Localization
 15. Install & run
 16. Development
 17. Caveats & limits
 18. Compatibility & mod coexistence
-

1. What it does (player-facing)

Each of your turns, the mod looks at every city in the world it can see and moves population from the least desirable settlements toward the most desirable ones:

- **Peacetime, unhappiness-driven.** Happiness is the single biggest factor, so an unhappy or low-yield city steadily loses population to happier, wealthier ones (no war required).
- **War refugees.** A city under actual attack (its districts taking damage, or pillaging in its borders) sheds population fast, and refugees **flee away from the nearest invader**: an army pressing from the east drives people west.
- **Cross-civilization.** People flee from one civ to another, not just between your own cities. This is the mod's defining behavior.
- **Regional.** Migration is distance-penalized, so people move to *nearby* better settlements rather than teleporting across the map.
- **Consequential.** Receiving migrants costs the destination civ happiness/gold for a while as it assimilates them, so a magnet city converges instead of accreting population forever. Hoarding unsettled migrant units costs too.

These layers, all on by default, add: refugees that **shun the aggressor** (own civ first, neutral third parties next, the attacker last); **border policies** — open up as a magnet, or seal off as a fortress that keeps your own people home and mobilizes a war economy (+**Production**) at the cost of **Influence**; **disasters** that push out climate/disaster refugees and a plague that migrants can carry; an **outlet** where a trapped, desperate population dies off rather than the system staying closed; and a nuanced happiness model that stops pure-happiness civs from becoming unstoppable magnets. All of it is reported in-game (toasts + the Demographics graphs) and in the dev log, e.g. **EMIGRATION 1 population point** (12 thousand people) left Rome (Romans) for Carthage (Carthaginians).

2. How it works: the per-turn loop

On every `PlayerTurnActivated`:

1. **Per-civ costs** (`chargePerTurnCosts`) run for *whichever* civ's turn it is: the decaying assimilation cost and the migrant-holding penalty (§7).
2. **The emigration pass** (`runPass`) runs once on the **local player's** turn (gated by `turnInterval`):
 1. **Decay** accumulated **violence** and **disaster** distress for the turn.
 2. **Collect signals**: one `CitySignal` per met city (§3).
 3. **Rank by Prosperity**: score every city; sort descending (§3, §5).
 4. **Advance state**: a monotonic turn counter (for scaling) + prune/tick cooldowns; compute per-owner populations (for congestion).
 5. **Process each source** until `maxMovesPerTurn` is hit:
 - Skip if its rural pool is at the floor or it's on cooldown.
 - Find its **best destination** (greatest *adjusted pull*, §3).
 - **If there's a destination**: accumulate **pressure**; on crossing `emigrationBar`, move one or more rural points depending on context (`addRuralPopulation(-1)` on the source, `+1` on the destination, works cross-civ). War-surge can raise same-turn war displacement up to `warSurgeMax` (still bounded by `siegeLossCapPct`), transit lag can delay arrival (`transitLagTurns/transitHexPerTurn`), and each successful intake records war-loss (when besieged), can seed plague carry (§6c), and adds **assimilation load** to the destination civ.
 - **If there's no destination** (the outlet, §6d): a sufficiently *distressed* source builds attrition pressure and eventually **loses a rural point with no destination**: population leaves the world (a death), not the city.
 6. **Persist** state to `GameConfiguration`; surface **feedback** (§9).
3. **Events**. Subscribed at boot: `DiplomacyDeclareWar/MakePeace` feed the aggressor map (§6a); `RandomEventOccurred` feeds disaster distress + a named alert (§6c).

3. The signals & the Prosperity score

`emigration-cities.js` builds a `CitySignal` per city (owner, population, rural pool, **urban** population, per-capita yields, net happiness, unrest, starvation, siege, war, accumulated **violence**, accumulated **disaster** distress, **infected** flag). `emigration-prosperity.js` turns it into a score. The **default (legacy linear)** model:

$$\begin{aligned}
 P &= (Q + h \lambda_h - n \lambda_n) \left(1 + \frac{s}{100}\right), \\
 Q &= \frac{f w_F + p w_P + g w_G + s c w_S + c w_C}{n}, \\
 s &= v + d + \sigma + \tau + u.
 \end{aligned}$$

Where P is prosperity, Q is per-capita productiveness, h is net happiness, n is population, and s is the summed situational percentage from violence, disaster, siege, starvation, and unrest channels.

Higher = more attractive. **Happiness dominates** (weight `localHappinessFactor`, default 6), which is why unhappiness drives migration even at peace; the situational multiplier is where war/violence, disasters, sieges, starvation, and unrest bite. (§5 replaces the happiness term and the violence penalty with more nuanced versions when their flags are on.) The magnitude of the negative situational percent is also exposed as **distress(s)**, which drives the outlet (§6d).

Violence (`emigration-violence.js`): polled, fog-independent

War-driven emigration keys on **actual violence inside a city's borders**, not on the empire merely being at war, and it's **symmetric** for player-watched and distant AI-vs-AI wars, because it reads game *state*, not visibility-gated events:

- **City under attack**: polls the city-center district's health (`Players.Districts.get(owner).getDistrictHealth/...MaxHealth`) readable for *all* players regardless of line of sight. Fresh damage spikes (`vwAssault`); standing damage sustains a siege (`vwSiege`).
- **Pillage**: damaged improvements on `getPurchasedPlots()` add standing pressure (`vwPillage`).
- The score **accumulates and decays** (`violenceDecay`): a sustained siege builds, a lone raid fades in 2-3 turns. With **Algorithm D** on, the curve also escalates with siege *duration* and is capped in total (§5-D).

Geography (emigration-geography.js)

- **Distance decay:** $-\text{distanceFactor} \times \text{hexDistance}$, keeping migration regional.
- **Flee-from-invader:** when violence crosses `violenceFleeThreshold`, refugees prefer destinations *away* from the nearest enemy (`fleeFactor`).
- **Aggressor preference:** own civ > neutral > the attacker, when Feature 1 is on (§6a).
- **Open Borders flow bonus:** a modest cross-civ pull bump between civs holding a base-game Open Borders agreement (`openBordersBonus`, §6b).

The destination decision (emigration-pull.js)

The destination scorer is implemented as two bounded channels over the prosperity gradient and friction terms:

$$\text{Pull}(s, d) = \left(\Delta\text{Pros}(s, d) + \text{Tilt}(s, d) - \text{Friction}(s, d) \right) \cdot \Pi(s, d),$$

$$\Delta\text{Pros}(s, d) = \text{Pros}(d) - \text{Pros}(s),$$

$$\text{Tilt}(s, d) = \text{clamp}(\text{asylumTilt}(s, d), -\text{tiltCap}, \text{tiltCap}),$$

$$\Pi(s, d) = \text{clamp}(\text{openness}(d) \cdot \text{retention}(s) \cdot \text{permOpenBorders}^{ob} \cdot \text{permAlly}^{al} \cdot \text{permWar}^{wa}, \text{permeFloor}, \text{permeCeil}).$$

In the permeability Π , $\text{openness}(d)$ is the **destination's** inbound border throttle and $\text{retention}(s)$ the **source's** cross-civ outbound throttle — the two halves of the Anti-Immigration stance (§6b). Both are 1 unless border policies are on, and retention applies only cross-civ (internal moves don't cost a civ population).

With friction:

```
$$ \begin{aligned} \mathrm{Friction}(s,d) = & \mathrm{baseReluctance} \\ & \cdot \mathrm{perExtraPop} \cdot \max\left(0, \mathrm{pop}(d) - \mathrm{pop}(s)\right) \\ & \cdot \mathrm{cityStateBarrier} \\ & \cdot \mathrm{poachBlock} \cdot \mathrm{geoAdjust}(s,d) \\ & \cdot \mathrm{congestionFor}(d). \end{aligned} $$
```

War is not a hard gate: a besieged source gets lower prosperity and directional flee pressure, then passes through the same pull equation. War is **not** a hard gate: a besieged city simply has low prosperity and a flee vector. People can emigrate to any civilization.

4. Population scaling (Demographics alignment)

`emigration-population.js` converts Civ's abstract population points into representative people using the **identical formula** to the Demographics mod:

```
scaleCityPopulation(raw, turn) = raw^scaleExp * scaleBase * scaleGrowth^turn // 1.11, 3000, 1.009
```

A moved point is reported as the **marginal** people it represents ($\text{scale}(\text{pop}) - \text{scale}(\text{pop}-1)$), using a **monotonic** turn so the figure never resets at age boundaries. `formatPeople` renders "12 thousand / 1.3 million / 240 million". `moveRural` performs a relocation; `removeRural` removes a point with no destination (the outlet's death, §6d), using the same rural-population accounting the game's own starvation shrinkage uses.

5. The advanced model (algorithms & per-civ tuning)

Four algorithms plus a per-civ tuning table refine the baseline, **all on by default** (each can be switched off in Options). Full math + before/after numbers: ../emigration-docs/algorithmic-improvements.md.

A. Shaped happiness (happinessShaped)

The linear **happiness** × 6 term let pure-happiness sources run away (Benjamin Franklin's Glass Armonica, +15 happiness/ally, made a ~50× magnet). The shaped model is **field-relative** (measured vs the world mean), **saturating** on the pull side and **steep** on the misery side (**tanh**), and makes happiness **amplify the economy** (bounded multiplier [happyMultMin, happyMultMax] + happyFloor) rather than dwarf it. Net: Franklin drops to ~2× while unhappy cities still shed strongly.

B. Overcrowding discount (overcrowdDiscount)

The probe (§12) confirmed population costs **zero** happiness per head; a tall city's unhappiness is **overcrowding** past a density threshold, and **getYield** is the **net, post-penalty** value, so unhappiness double-hits (suppressed yields *and* the ×6 term). The discount credits back density-driven unhappiness via **urbanPopulation** vs **overcrowdThreshold**.

C. Congestion headwind + leader variance (congestWeight)

A structural **anti-runaway brake that can't be out-golded**: a civ absorbing lots of migrants becomes a less attractive *further* destination, scaling with its per-capita assimilation load. Two leader-variance knobs ride the assimilation cost via the civ table: **integrationSpeed** (load decay) and **assimilationEase** (gold cost).

D. Capped, time-gated war displacement (warSiege)

Fog-independent violence made war a bloodless depopulation tool. The siege model tracks **siege tenure**, **escalates** the penalty from **siegeFloor** to full over **siegeRampTurns**, and **caps** total war loss at **siegeLossCapPct** of onset population (the remnant "digs in"), so a city can be gutted but never emptied without a capture.

The civ tuning table (emigration-civ-tuning.js, civTuningEnabled)

A small, auditable registry of **bounded** per-leader/per-civ nudges, keyed on the GameInfo leader string (_ALT personas normalized; leader overrides civ). Fields: **happinessPull**, **integrationSpeed**, **assimilationEase**, **overcrowdDiscount**, **warRetention**, **sourceBias**. Shipped entries target outliers: Franklin **happinessPull** 0.75, Isabella **0.85+ease** 1.2, Xerxes **ease** 1.25, Khmer **sourceBias** 1.5, Pachacuti **overcrowdDiscount** 0.5, Norman/England **warRetention** 1.4, and so on. None can cause a runaway; the structural guarantees live in the algorithms.

6. Interactive systems (on by default)

6a. Aggressor-aware war refugees (aggressorPenalty, 0 = off)

When civ A attacks civ B, B's refugees prefer **B's own cities** first, then **any civ other than A**, and treat **A** as a last resort. The aggressor is read from the public **DiplomacyDeclareWar** event (**actingPlayer** declared on **reactingPlayer**, probe-confirmed field names), persisted as a victim→aggressors map in **emigration-war.js** and cleared on peace. The preference (**ownCivRefugeeBonus** toward own civ, **-aggressorPenalty** for the attacker) is folded into **geoAdjust** only for cities actually under violence.

6b. Immigration-stance policies + Open Borders agreements

Two distinct levers control cross-civ immigration.

Your stance (a policy card, bordersEnabled). Slot **Pro-Immigration Stance** or **Anti-Immigration Stance** - renamed from "Open/Closed Borders" so they don't collide with the base game's Open Borders *diplomatic agreement*. A small **database component** (**data/emigration-policies-{antiquity,exploration,modern}.xml**, one file per age) adds the slot-able traditions, one per age, available to every civ. They unlock from a **mid-age** civic node (Antiquity **Citizenship**, Exploration **Economics**, Modern **Social Question** - roughly halfway through each age, all thematically on-point) so the cards arrive when migration is actually in play, rather than sitting useless in a policy slot from turn one. (Internal trait IDs keep **TRADITION_EMIG_OPEN/CLOSED_BORDERS_***.) The two stances are deliberately **asymmetric** — Pro is a growth/magnet play, Anti is a *fortress* play, not just its negation:

- **Pro-Immigration Stance:** +50% immigration **into** your cities (`immigrationOpenness(destOwner)`, a custom UI-VM multiplier on inbound pull) plus a native **+1/+2/+3 Influence TraditionModifier** for your cosmopolitan reputation.
- **Anti-Immigration Stance:** a fortress with **four** effects. It throttles inbound immigration to 40% (`immigrationOpenness`, floored at 0.15) **and retains your own people** — your citizens' *cross-civ* outbound pull is cut to 60% (`emigrationRetention(srcOwner)`, the mirror of the inbound throttle; cross-civ only, since internal moves don't cost your civ population). To offset the diplomatic cost of isolation it also rides a native **+2/+3/+4 Production in every city** modifier — a mobilized, autarkic war economy — alongside the native **−2/−3/−4 Influence** penalty (closing your borders is not internationally popular).

The migration % and retention are custom UI-VM mechanics; the Influence and Production are native **TraditionModifiers** (`data/emigration-policies-gameeffects.xml`), so they show on the card *and* in the yields breakdown like any base-game policy.

Diplomatic Open Borders (a flow bonus, `openBordersBonus`). When two civs hold an active base-game **Open Borders** agreement, migration between them is eased both ways: `emigration-geography.js` checks the joint diplomatic events (`Game.Diplomacy.getJointEvents(...).actionTypeName === "DIPLOMACY_ACTION_OPEN_BORDERS"`) and adds a modest cross-civ pull bonus. Console check: `emigration.openBorders(aPid, bPid)`.

Governments no longer separately affect emigration.

Policy cards by age (what is actually shipped) The policy layer is broader than the two immigration stances. The card set is age-scoped in `data/emigration-policies-{antiquity,exploration,modern}.xml`, with native per-turn yield effects in `data/emigration-policies-gameeffects.xml`. Card names below are the **in-game display names**; the internal trait tokens (which keep the OPEN/CLOSED_BORDERS wording) are listed in the ID reference that follows.

Antiquity cards

Policy (Civic)	Effects (native + migration)
Pro-Immigration Stance(Citizenship)	+1 Influence/turn Migration: pull x 1.5(<code>openBordersOpenness</code>)
Anti-Immigration Stance(Citizenship)	-2 Influence/turn, +2 Production/city Migration: inbound pull x 0.4 (floor 0.15);own cross-civ outbound x 0.6 (retention)

Exploration cards

Policy (Civic)	Effects (native + migration)
Pro-Immigration Stance(Economics)	+2 Influence/turn Migration: pull x 1.5(<code>openBordersOpenness</code>)
Anti-Immigration Stance(Economics)	-3 Influence/turn, +3 Production/city Migration: inbound pull x 0.4 (floor 0.15);own cross-civ outbound x 0.6 (retention)
Talent Attraction(Inspiration)	+1 Science/turn Migration: +1.5 Science poolper arrival
Cultural Magnetism(Society)	+1 Culture/turn Migration: +1.5 Culture poolper arrival
Commercial Draw(Mercantilism)	+1 Gold/turn Migration: +1.5 Gold poolper arrival
Selective Asylum(Piety)	+1 Influence/turn Migration: refugee pull tilt

Modern cards

Policy (Civic)	Effects (native + migration)
Pro-Immigration Stance(Social Question)	+3 Influence/turn Migration: pull x 1.5(<code>openBordersOpenness</code>)
Anti-Immigration Stance(Social Question)	-4 Influence/turn, +4 Production/city Migration: inbound pull x 0.4 (floor 0.15);own cross-civ outbound x 0.6 (retention)
Talent Attraction(Modernization)	+2 Science/turn Migration: +1.5 Science poolper arrival
Cultural Magnetism(Natural History)	+2 Culture/turn Migration: +1.5 Culture poolper arrival
Commercial Draw(Capitalism)	+2 Gold/turn Migration: +1.5 Gold poolper arrival

Policy (Civic)	Effects (native + migration)
Refugee Compact(Political Theory)	+2 Influence/turn, +1 Culture/turn Migration: refugee pull tilt

Internal ID reference (token → in-game name: OPEN_BORDERS = Pro-Immigration Stance, CLOSED_BORDERS = Anti-Immigration Stance, TALENT = Talent Attraction, CULTPULL = Cultural Magnetism, TRADEPULL = Commercial Draw, ASYLUM = Selective Asylum / Refugee Compact)

- Prefix: TRADITION_EMIG_
- Antiquity: tokens OPEN_BORDERS, CLOSED_BORDERS; suffix _ANTIQUITY
- Exploration: tokens OPEN_BORDERS, CLOSED_BORDERS, TALENT, CULTPULL, TRADEPULL, ASYLUM; suffix _EXPLORATION
- Modern: tokens OPEN_BORDERS, CLOSED_BORDERS, TALENT, CULTPULL, TRADEPULL, ASYLUM; suffix _MODERN

Native card yields are visible in the game breakdown. Migration-specific percentage and targeted-attraction behavior remains in UI-VM logic (`emigration-borders.js`, `emigration-pull.js`).

How attraction policy card yields actually function Attraction cards have two yield layers, and they stack:

1. **Native fixed yield from the DB card itself** (`data/emigration-policies-gameeffects.xml`): a constant per-turn yield shown in the normal game breakdown (for example, Talent gives Science, CultPull gives Culture, TradePull gives Gold; values scale by age).
2. **Carried dividend from actual immigrant intake** (`emigration-dividend.js`): each incoming migrant under an active attraction adds pool:

$$\text{pool}_y \leftarrow \text{pool}_y + \text{dividendPerMigrant}$$

Each turn, that pool decays and grants capped yield:

$$\begin{aligned} \text{pool}_y &\leftarrow \text{pool}_y \cdot \text{dividendDecay}^{\Delta t}, \\ \text{grant}_y &= \min(\text{dividendCap}, \text{pool}_y) \end{aligned}$$

So the card is not only a flat bonus; it also converts migration throughput into ongoing yield. This is why attraction cards feel stronger in high-intake destinations.

Default constants are: `dividendPerMigrant` = 1.5, `dividendDecay` = 0.7, and `dividendCap` = 12 per turn in a yield channel.

Important: +2 Influence/turn is **not** per immigrant. It is a flat per-turn civ yield modifier from the slotted card. The per-immigrant part is only the carried-dividend pool for attraction cards, applied on each successful arrival.

6c. Environmental disasters & plague (`disastersEnabled`)

Civ VII's `RandomEvents` (flood / volcano / **plague** / hurricane / blizzard / tornado / duststorm / thunderstorm) become a migration driver, parallel to war. `emigration-disasters.js` accumulates per-city **disaster distress** that decays each turn and feeds a situational prosperity penalty, so struck cities shed **climate/disaster refugees**. It's **fog-independent**: the canonical signal is `city.isInfected` (the base game's outbreak flag, which already makes a city emit migrants) plus a severity-scaled spike from the `RandomEventOccurred` event. **Plague-as-contagion** (`plagueCarryEnabled`, off): migrants fleeing an infected city seed a smaller outbreak-distress at their destination, a contagion vector beyond the game's own trade-network spread.

6d. The outlet: attrition when there's no refuge (`attritionEnabled`, off)

Keeps the model from being a **closed system**. When a source has **no viable destination** *and* is genuinely distressed (`distress(s) >= attritionMinDistress`, siege/starvation/heavy-violence/disaster level), it builds attrition pressure and, on crossing `attritionThreshold`, loses a **rural point with no destination**: population leaves the *world* (a

death/dispersal), via the same `addRuralPopulation(-1)` accounting the game's starvation uses. A content city with nowhere better to go is untouched. Attrition is tracked as **deaths**, fully isolated from the migration/refugee metrics: a death isn't a migration, so it never inflates any flow figure.

6e. Asylum and relationship permeability (targeted attraction)

Pull is composed from a prosperity gradient plus a targeted-attraction channel (`tilt`) and relationship permeability multipliers. In practice:

- **Asylum push** (`asylumPushWeight`) adds targeted pull for distressed refugees toward destinations that are actively hospitable.
- **Relationship permeability** (`permOpenBorders`, `permAlly`, `permWar`) scales cross-civ movement without hard-forcing it.
- **Global bounds** (`tiltCap`, `permeFloor`, `permeCeil`) prevent runaway attraction or hard lockout.

This logic is computed in `emigration-pull.js`, so targeted attraction composes with the rest of the destination decision rather than bypassing prosperity/geography/congestion.

7. Consequences: the gameplay-write cost layer (`emigration-effects.js`)

Civ VII makes raw population *free*, so the mod adds the missing feedback via `Players.grantYield(pid, YIELD_X, -amount)` (probe-confirmed to deduct, cross-civ; happiness leg inferred):

- **Assimilation cost (duration-based)**. Each migrant adds **load** to the receiving civ (`assimilationLoadPerMigrant * (1 + assimilationCostPerPop * destPop)`). Load **decays each turn** (`assimilationDecay`, the duration, optionally scaled by `integrationSpeed`) and the civ pays per-turn `assimilationHappiness/assimilationGold` per unit (gold leg optionally scaled by `assimilationEase`). Scoped to *migrated* population only.
- **Migrant-holding penalty**. Per-turn cost per unsettled `UNIT_MIGRANT` a civ holds, so overflow migrants must be settled, not hoarded.
- **Congestion headwind (Algorithm C)**. `congestionPenalty + assimLoadFor` live here; the engine subtracts the headwind from a destination's pull.
- **Carried dividend (the assimilation mirror)**. Under attraction contexts, incoming migrants can build a decaying per-turn positive pool (`dividendPerMigrant`, `dividendDecay`, `dividendCap`) in the matched yield domain (science/culture/gold). Implemented in `emigration-dividend.js` and granted through the same verified `grantYield` surface.

All apply to **every civ on its own turn**. Set any knob to 0 to disable.

8. Reporting & Demographics integration (`emigration-migration-stats.js`)

When the **Demographics** mod is installed, Emigration contributes, via its companion hook (`globalThis.DemographicsMetricsAPI`, an order-independent handshake):

- **Graphs on the Power page: Net migration, Emigration** (gross out), **Immigration** (gross in), each per civ over time, in the same scaled people units as Population.
- **Per-cause source attribution** on the Emigration/Immigration graphs: when Demographics supports tooltip registration (`registerMetricTooltip`), each graph carries a **Sources: War: ..., Disaster: ..., Attraction: ..., Unhappiness: ...** breakdown derived from the per-cause tallies. It degrades gracefully (no tooltip) on an older Demographics that lacks the hook.
- **A dedicated Migration page** on the Demographics screen (when its `registerPanel` companion hook is present): the same dashboard as the standalone window — per-civ ledger, the per-cause breakdown, border stances, and your cities under pressure. Registered order-independently and a silent no-op on an older Demographics (the standalone window still covers it).
- **A Refugees graph** on the Conflicts page, and a **Refugees row in the war-effects tooltip** (a small Demographics-side edit: `COST_METRICS` + the war sampler reading

`globalThis.EmigrationData.refugeesCumFor`). Refugees attribute to the besieged side and render "- no data" when Emigration isn't installed.

`EmigrationData` (exposed globally) carries per-civ cumulative tallies: gross in/out, net, **refugees** (war/disaster/conquest displacement), **deaths** (attrition), and the **per-cause** emigration/immigration breakdowns (`emigrationByCauseFor`, `immigrationByCauseFor`). If Demographics isn't installed it's all a silent no-op.

9. In-game feedback & notifications (`emigration-feedback.js`, `emigration-events.js`, `emigration-naming.js`)

Migration is surfaced as **styled HUD toasts** (matching the game's look: `BodyFont`, the dark panel tones, parchment text) and, for big events anywhere in the world, **world-news**. It is **important-only by design**, with several anti-spam layers (the engine has no toast API and `WorldUI` has no floating-text method, so toasts are the channel):

- **Rich, named events** (`emigration-naming.js`, modeled on the Demographics war-naming): disasters use the **game's own names** (`GameInfo.RandomEvents.lookup(type).Name`, e.g. "Thera" or "Catastrophic Eruption"), wars reuse the war name, conquest names the sacked city. Headlines like *"The Thera eruption displaces 80,000."*
- **Explanatory & actionable** (the in-game-legibility layer). When *your* cities lose people in a pass, a single throttled **local digest** toast answers why / what-to-do / temporary-or-permanent / who-pays — e.g. *"12 thousand people left Rome, unhappy at home. Raise this city's happiness, or slot an Anti-Immigration Stance to retain them. It continues until you address the cause."* The cause-keyed **action hint** (`emigration-causes.js#causeHint`) and **permanence cue** also ride the verbose per-cause toasts and the disaster alert. It uses the same important-toast cooldown, so richer content does **not** mean more toasts.
- **Per-city readout** (`emigration-city-readout.js`). An on-demand HUD panel answering "why is *this* settlement changing?" — dominant cause + status (building pressure / resting), where its people are pulled, the assimilation cost, the civ's net migration, the hint, and an at-risk / trapped-with-no-refuge warning. Built from the recomputed `citySnapshot` (no new state); opens via `emigration.city(id)` / `emigration.hideCity()` and best-effort on city selection. Toggle in Options (`cityReadoutEnabled`); works without Demographics.
- **Dashboard window** (`emigration-window.js` + the shared render core `emigration-views.js`). A standalone HUD window (`emigration.window()` / `emigration.closeWindow()`) with the whole picture: a per-civ ledger (in/out/net/refugees/deaths), the "why people move" cause breakdown, who holds Pro-/Anti-Immigration stances, and your cities ranked by migration pressure. The same render core will back the Demographics page, so it works either way.
- **Anti-spam**. Disasters only notify at/above `disasterNotifyMinSeverity` (minor events drive the sim silently). War notifications are **once-per-milestone** on a civ's **cumulative** refugees (`worldRefugeeThreshold`), not per turn. A global `notifyCooldownTurns` backstops everything. `notifyMode`: **0** off / **1** important-only (default) / **2** verbose (adds a per-pass per-cause toast). All player-tunable.
- **Events**. `RandomEventOccurred` → disaster distress + a named alert; war/peace events feed the aggressor map.

10. Options & tuning

Everything is under **Options** → **Mods**, in **both** the main-menu (pregame) and in-game Options screens. `emigration.modinfo` loads the options layer in both shell and game scopes, and it registers via `Options.addOption({category: CategoryType.Mods, ...})`. Settings persist in the shared, cascade-safe `modSettings` `localStorage` slice and apply to the live config immediately and at game boot.

- **Emigration group**: **Migration counts** (Both / Civ only / Historical only) and **Emigration intensity** (Custom / Low / Medium / High).
- **Emigration - Advanced**: every tunable as a dropdown/checkbox, generated from a declarative spec (`emigration-tunables.js`), grouped: pacing, scope, prosperity weights, the **advanced-model** switches (§5: shaped happiness, overcrowding, congestion, civ tuning), **war/violence** (incl. the siege model + aggressor avoidance), **border policies**, **geography**, assimilation/migrant **cost**, **disasters** (+ plague carry), **notifications** (verbosity, min severity, cooldown, crisis threshold), and the **outlet** (attrition).

The full default set lives in `emigration-config.js`; scaling constants are intentionally **not** exposed (they must match Demographics).

11. Architecture / module map

Modules are small and single-concern (the repo enforces a ≤ 500 -line file gate), so several systems span a parent plus split-out helpers.

- `ui/emigration-main.js`: Entry UI Script with per-turn hook/costs, event subscriptions, reporting/feedback orchestration, dev dock, and boot.
- `ui/emigration-config.js`: Tunable default values, `CONFIG_DEFAULTS` snapshot, and scaling constants.
- `ui/emigration-config-types.js`: `EmigrationConfig` typedef schema for `CONFIG`.
- `ui/emigration-causes.js`: the single source-of-truth migration-cause taxonomy (one `MigrationCause` typedef + `causeLabel` / `causePermanence` / `causeHint` / `isRefugeeCause`).
- `ui/emigration-tunables.js`: Declarative exposed knobs plus Low/Med/High presets.
- `ui/emigration-cities.js`: Enumerates met cities into `CitySignal` records.
- `ui/emigration-prosperity.js`: Prosperity scoring (legacy + shaped happiness + overcrowding) and `distress`.
- `ui/emigration-violence.js`: Violence state machine (accumulate/decay) plus siege tenure/escalation/cap.
- `ui/emigration-violence-signals.js`: Fog-independent polled combat signals (`district damage`, `pillage`).
- `ui/emigration-disasters.js`: Per-city disaster distress, decay, and plague-carry seeding.
- `ui/emigration-geography.js`: Distance decay, flee-from-invader, aggressor preference, and Open Borders flow bonus.
- `ui/emigration-civ-tuning.js`: Per-leader/civ tuning table and resolver.
- `ui/emigration-war.js`: Aggressor map from `DiplomacyDeclareWar/MakePeace`.
- `ui/emigration-borders.js`: Border-policy reads $\rightarrow$ `immigrationOpenness` (inbound) + `emigrationRetention` (outbound) multipliers, attraction yields, asylum flag.
- `ui/emigration-effects.js`: Assimilation cost load (+ leader variance) and congestion headwind.
- `ui/emigration-dividend.js`: Carried dividend attraction benefit per immigrant.
- `ui/emigration-migrant-units.js`: Per-turn penalty for unsettled `UNIT_MIGRANT` units.
- `ui/emigration-engine.js`: Main pass (ranking, pressure, move, and outlet).
- `ui/emigration-pull.js`: Destination decision (`adjustedPull`, `bestDestination`, `migrationCause`).
- `ui/emigration-state.js`: Engine-state persistence (pressure, cooldowns, scaling turn, per-owner populations).
- `ui/emigration-population.js`: Population reads/writes and Demographics scaling.
- `ui/emigration-migration-stats.js`: Per-civ tallies, the session-local recent-moves feed, and the `EmigrationData` global.
- `ui/emigration-city-readout-data.js`: the per-city "why is this settlement changing?" snapshot — a pure `buildCitySnapshot` view-model builder plus a `recompute-on-read citySnapshot(cityId)`.
- `ui/emigration-city-readout.js`: the on-demand per-city readout panel — a pure `readoutModel` plus a HUD-anchored DOM host (`showCityReadout` / `hideCityReadout`) and selection wiring.
- `ui/emigration-views.js`: the shared dashboard render core — pure view-model builders (civ ledger, per-cause breakdown, border stances, per-city pressure table) + `renderDashboard`.
- `ui/emigration-window.js`: the standalone dashboard window host — gathers world migration state (`gatherDashboard`) and mounts the render core (`emigration.window()` / `emigration.closeWindow()`).
- `ui/emigration-migration-page.js`: registers a dedicated Migration page on the Demographics screen via its `registerPanel` hook (order-independent; no-op if unsupported), mounting the shared render core.
- `ui/emigration-demographics.js`: Demographics graph registration (Net/Out/In/Refugees), incl. the per-cause source-attribution tooltips on the Emigration/Immigration graphs.
- `ui/emigration-demographics-per-cause-metrics.js`: optional, **not shipped/loaded** — an example that registers one line chart per cause (war/disaster/prosperity/unhappiness) for mods that want granular per-cause graphs instead of the tooltip breakdown.
- `ui/emigration-naming.js`: Rich event naming.
- `ui/emigration-feedback.js`: Styled toasts, world-news, and anti-spam throttling.
- `ui/emigration-report.js`: Renders `Migration` records into log lines.
- `ui/emigration-log.js`: Dev debug logging via `UI.log` CSS-parse channel.
- `ui/emigration-events.js`: `RandomEventOccurred` $\rightarrow$ disaster distress + named alert.
- `ui/emigration-settings.js`: Number-display preference and tunable/preset getters/setters.
- `ui/emigration-options.js`: Registers presets and Advanced tunables under Mods options.
- `ui/options/emigration-advanced-editor.js`: Advanced tunables sub-window screen.

- `ui/options/mod-options.js`: Shared Mods category and cascade-safe `modSettings` store.
- `data/emigration-policies-{antiquity,exploration,modern}.xml`: Database components for stance policy cards/unlocks (one file per age).
- `data/emigration-policies-gameeffects.xml`: Database component for native `TraditionModifier` Influence/yield effects.
- `data/emigration-policy-icons.xml`: Icon database component for stance/attraction cards.
- `data/emigration-civilopedia.xml`: Database component for Emigration Civilopedia pages.
- `ui/migration-probe.js`: Dev-only API probe module (separate modinfo, not shipped).
- `text/<locale>/ModText.xml`: Localized strings for all 10 locales (generated; see §14).
- `scripts/i18n_extract.mjs` and `scripts/i18n_apply.mjs`: Dev-only localization pipeline.

`emigration.modinfo`'s `ActionGroups`: the options layer in **both** shell + game scopes, the engine and its sub-modules (`ImportFiles`) in game scope, an always-on `<UpdateDatabase>` for the Civilopedia pages + the policy `TraditionModifiers`, an `<UpdateIcons>` for the card art, and **three age-scoped** `<UpdateDatabase>` groups for the border policies (one per age via `AgeInUse` criteria). The per-age split is required: Civ VII rebuilds the gameplay database each age, so a policy's civic-tree unlock can only reference nodes that exist in that age's database (loading the Exploration/Modern unlocks into the Antiquity database is a foreign-key failure). UI modules are ES imports resolved by path from the deployed file tree; the `ImportFiles` list is kept a complete inventory of that tree and is gated by a manifest test (`tests/modinfo.mjs`) so a future module split can't silently leave a file unlisted.

The Civilopedia component adds an **Emigration** section to the in-game encyclopedia with an overview plus one page per system (Prosperity, War & Refugees, Assimilation, Borders & Influence, Disasters & Plague, the Outlet, and Leaders & Civilizations, the per-civ tuning), reusing the base `Concept` page layout. The stance policy cards get their own `pedia` pages automatically from their tradition Name/Description, so they aren't duplicated. All page text flows through the same localization pipeline (§14): `LOC_PEDIA_EMIG_*` keys, all 10 locales.

12. The execution model & what the probe confirmed

The mod runs in the **UI VM** (GameFace JS), far more capable than "UI only", verified in-engine with the `migration-probe` dev mod (its `API3` + `API4` button passes + passive event recorders):

- `city.addRuralPopulation(±1)`: the population move/removal. Not owner-gated → works cross-civ. (The only population write, which is why the outlet "kills" via the same channel as starvation.)
- `Players.grantYield(pid, YIELD_X, ±n)`: yield costs; not owner-gated, **negative deducts** (gold confirmed, cross-civ).
- `DiplomacyTreasury.changeDiplomacyBalance(±n)`: Influence write, confirmed cross-civ. (Superseded - border-policy Influence is now a native `TraditionModifier`, so the mod no longer writes Influence directly.)
- **War aggressor**: `DiplomacyDeclareWar` carries `actingPlayer` (declarer) + `reactingPlayer` (target).
- **Reads (fog-independent)**: `district.health`, `city.isInfected`, `Culture.getActiveTraditions` / `isTraditionActive`, `GameInfo.RandomEvents` (class/severity/name), `player.leaderType/civilizationType`, and a cheap per-civ population aggregate, all read for *all* players.
- **The happiness economy**: `pop upkeep happiness = 0`, `OVERCROWDING_THRESHOLD = 2`, `getYield === getNetYield` (net/post-penalty), the grounding for Algorithm B.

What the VM **cannot** do (and the mod doesn't): create units for *other* civs (`CREATE_ELEMENT` is local-only), or raise a custom **notification type** without a DB entry (so engine notifications are deferred; toasts + world-news cover feedback). The policy *cards* are the one piece that needs the database; everything else is the direct-mutator surface.

13. Persistence

Per-game state lives in the `GameConfiguration` KV store (survives save/reload):

- `EmigrationState_v1`: per-source pressure, cooldowns, the monotonic scaling turn.
- `EmigrationViolence_v2`: per-city violence intensity + decay, and siege tenure / onset population / cumulative war-loss.
- `EmigrationDisaster_v1`: per-city disaster distress + decay.

- **EmigrationAssim_v1**: per-civ assimilation load + per-civ tick turn.
- **EmigrationWar_v1**: victim → aggressors map (Feature 1).
- **EmigrationMigStats_v1**: per-civ tallies: net, gross in/out, refugees, deaths, plus the per-cause emigration/immigration breakdowns and their graph-sample watermarks.
- **EmigrationNews_v1**: world-news announced-milestone tiers + the last-toast turn (anti-spam).

Options/settings persist separately in the shared `modSettings` `localStorage` key (never a stray top-level key). Single-player scope (UI-VM gameplay writes are client-side).

14. Localization

All user-facing strings are LOC keys, **fully translated into all 10 locales** (en, de, es, fr, it, ja, ko, pt, ru, zh), including the advanced tunable labels. The non-English files are **generated, not hand-edited**:

```
node scripts/i18n_extract.mjs    # text/en_us/ModText.xml → i18n/i18n-source.json (key list)
node scripts/i18n_apply.mjs      # i18n/<locale>.json → text/<locale>/ModText.xml
```

Author English in `text/en_us/ModText.xml`; translations live in `i18n/<locale>.json` (a missing key falls back to English). `npm run verify` includes a **parity gate** (`tests/i18n.mjs`) that fails if any `en_us` key is absent from a locale, so nothing silently drifts. `{1...}` placeholders and code tokens (`UNIT_MIGRANT`, `Demographics`) are preserved verbatim. (Machine-quality translations; a native-speaker review pass is welcome and is a one-line re-apply.)

15. Install & run

1. Copy this folder to `~/Library/Application Support/Civilization VII/Mods/emigration/`, relaunch, and enable **Emigration** in *Additional Content*.
2. Play turns. With `const DBG = true` (dev default) the mod logs to `UI.log` via the CSS-parse channel (mod `console.log` doesn't reach the log files):

```
grep -E "EMIG_" "~/Library/Application Support/Civilization VII/Logs/UI.log"
```

`release.sh` flips `DBG` to `false`, so shipped builds run silently.
3. **Dev dock buttons** (subsystem dock): run a pass now / dump the prosperity ranking. Console: `emigration.runNow()`, `emigration.rank()`.
4. **Tune or disable the layers**: Options → Mods → Emigration - Advanced exposes every advanced-model switch (§5) and interactive system (§6: aggressor avoidance, border policies, disasters, the outlet), plus the notification mode. All default **on**; switch any off to walk back toward the bare baseline.

Look for **EMIGRATION ... left ... for ...**, **assimilation: ... cost lines**, and **ATTRITION ... (no refuge)** when the outlet fires.

16. Development

Typed JavaScript with JSDoc, no build step: what ships is what you write (see `CONTRIBUTING.md`). Before committing:

```
npm install
npm run verify    # tsc --noEmit + eslint (0 warnings) + the node test harnesses
```

`verify` runs the modularization gate (file/function length / complexity / statements) and **22** test harnesses: **causes** (cause taxonomy: labels, permanence, refugee routing), **city-readout-data** (per-city snapshot view-model), **city-readout** (readout panel model), **views** (dashboard render-core builders), **migration-page** (Demographics page handshake), **scaling**, **prosperity** (shaped happiness + overcrowding + distress), **geography**, **violence** (siege escalation + cap), **tunables**, **migration-stats** (gross / refugees / deaths / per-cause / recent feed), **effects** (congestion + leader variance), **engine-pull** (destination decision), **dividend** (carried-dividend grants), **civ-tuning**, **war** (aggressor map + ordering), **disasters** (distress + decay + severity), **borders** (openness + influence), **naming** (headline fallbacks), **feedback** (anti-spam throttling), **modinfo** (the `ImportFiles` manifest stays a complete, import-closed inventory), and

i18n (locale parity). `./release.sh` produces the debug-muted, allow-listed Workshop zip (readable JS, no minification; `data/` and the 10 locales included, dev cruft excluded).

The **migration-probe** mod (its own modinfo, never shipped) is the in-engine verifier behind the write-surface/data claims: dock buttons + a `globalThis.mig` console API, with the **API3** (identity, specialist/overcrowding semantics, yield pre/post-penalty, per-civ aggregate) and **API4** (influence write, government/policy reads, `isInfected`, `RandomEvents`, notification surface) confirmation passes + passive `DiplomacyDeclareWar` / `RandomEventOccurred` recorders.

17. Caveats & limits

- **Single-player.** UI-VM gameplay writes are client-side.
- **The advanced layers are on by default but un-playtested-at-scale.** Everything beyond the baseline is implemented and unit-tested, but the knob values are starting points to tune against real games, not balanced settings. Turn pieces off in Options (§5-§6) if a save hits balance trouble or you want the bare baseline.
- **Happiness cost is inferred.** Negative *gold* grants are probe-confirmed; negative *happiness* is inferred; set the happiness knobs to 0 if a build no-ops them (the congestion headwind, §5-C, is the gold-immune structural brake).
- **A few in-engine confirmations remain best-effort:** the policy cards' in-game slotting/unlock, whether disaster *plot-effects* are pollable per plot (we use `isInfected`, which is confirmed), and a longitudinal `getYield` pre/post-penalty check. The code degrades to a safe no-op where unconfirmed.
- **On-map floating indicators are deferred:** `WorldUI` exposes no floating-text method (only overlay/marker builders), so feedback uses toasts; a built overlay is a future add.
- **Engine notifications need a DB type:** the clickable end-turn notifications would need a `NotificationType` (a future addition alongside the policy data); toasts + world-news cover it for now.
- **No new game rules.** The mod composes existing engine writes; it can't invent effects or spawn units for the AI.

18. Compatibility & mod coexistence

The mod is built to share the game with others. Its only dependency is **base-standard**, and it touches every shared surface *additively*:

- **Database: inserts only.** `data/emigration-policies-*.xml` and `data/emigration-civilopedia.xml` are pure `<Row>` inserts: no `<Replace>`, `<Update>`, or `<Delete>` against base or shared tables, so the mod can't overwrite another mod's content. New IDs are namespaced (`TRADITION_EMIG_*`, `Civilopedia SectionID="EMIGRATION" / PageGroupID="EMIG_TOPICS"`). The border-policy traditions attach to existing civic-tree nodes via additive `ProgressionTreeNodeUnlocks` rows without modifying the nodes, and the Civilopedia pages reuse the base Concept layout read-only.
- **Shared settings store, self-healing.** Options persist under the single community-convention `modSettings` `localStorage` key (§13), never a stray top-level key. Because GameFace's `localStorage` is shared across *every* mod, a stray top-level key silently clobbers that store and breaks settings for all installed mods, so on each save the store self-heals, preserving `modSettings` and dropping stray keys (`ui/options/mod-options.js`). With any mod that follows the convention (Demographics does) this is a no-op and they coexist in one object; the only time it touches another mod's storage is to clear a key that is already breaking the shared store for everyone.
- **Cooperative globals + events.** JS globals are namespaced (`globalThis.emigration`, `EmigrationData`). The Demographics integration uses an order-independent handshake (`globalThis.DemographicsMetricsAPI ??= {}`) that *joins* the metrics API rather than replacing it, with registration queued so load order doesn't matter. Engine events (`engine.on(...)`) are multicast, so subscribing to `PlayerTurnActivated` / `DiplomacyDeclareWar` / `RandomEventOccurred` never blocks another mod's handlers.
- **Defers to the Demographics namespace.** The war-popup refugees label and glossary are owned by the Demographics mod; Emigration adds only the one graph-title string it introduces (`LOC_DEMOGRAPHICS_WAR_GRAPH_T_REFUGEES`), so there are no duplicate `LocalizedString` definitions.
- **Adaptive to other mods.** It reads *live* happiness / yields / Influence each turn, so a mod that rebalances those values is simply reflected in the Prosperity score, so effects compose rather than conflict. At the gameplay level it shares the world: if another mod also moves population or changes yields, the two stack without crashing or corrupting state. (The advanced layers default on; turn any off in Options, §5-§6, to coexist more conservatively.)

Credits & license

Adapts the *Prosperity* data model from Machiavelli's "Emigration" mod for Civilization V. MIT.

See LICENSE.